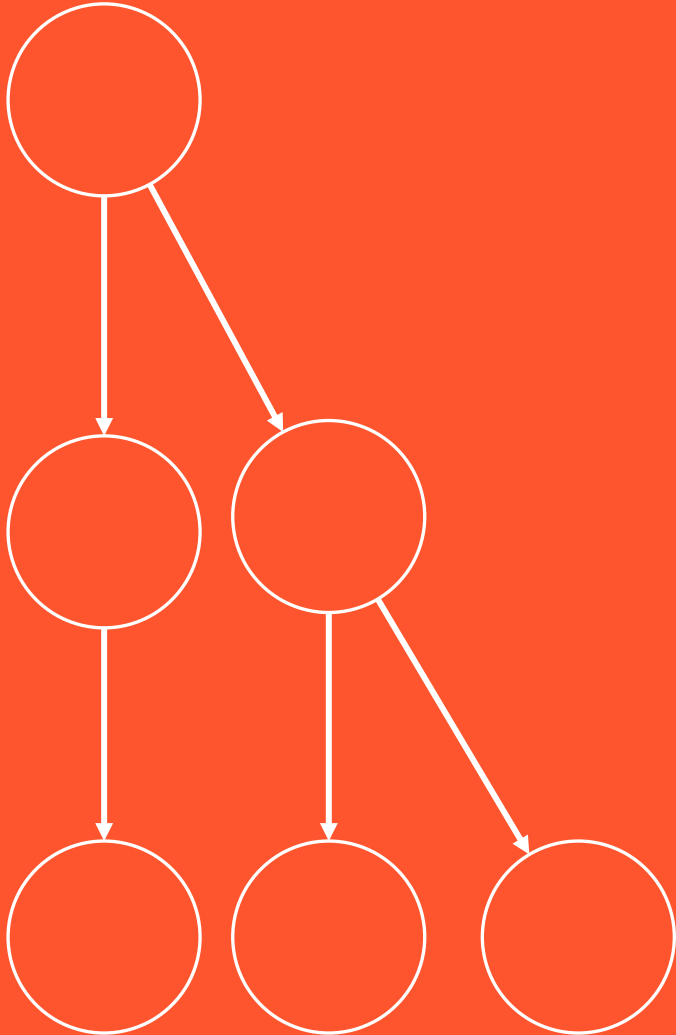


BCOG 100

Comparative Approaches

Lecture 2.3 — Comparing Minds, Machines, and Methods



UNIVERSITY OF
ILLINOIS
URBANA - CHAMPAIGN

Three Questions



Can anything but a human brain have a mind?
Can a machine?

We gave a reason to think the question is legitimate:
a mind looks like a property of how a system is organized, not of what it happens to be made of.

If that is right, nothing rules out a machine having one in advance.

It also does not settle that a machine does.

The question gets no clean yes or no.

People asking whether a machine *really* thinks are usually asking **three different questions at once**, without noticing.

Most disagreements about artificial intelligence are disagreements about **which of the three is being asked.**

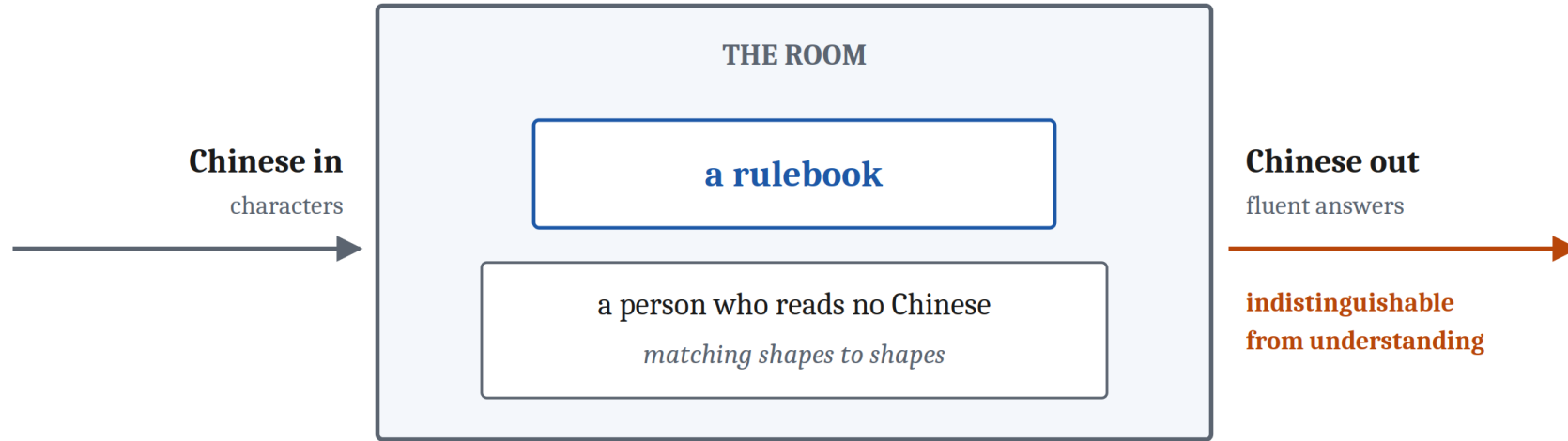
Three Ways a Machine Can Match a Mind

Three senses, and they license completely different conclusions.

	What is claimed	What would establish it	What it licenses
Behavioral equivalence	The two systems do the same things, judged from outside	The same outputs for the same inputs — including where each fails, and how each degrades under pressure	What the system does. Nothing about how
Mechanistic equivalence	They reach those outputs the same way	The same operations over the same kinds of intermediate representation, shown by evidence from inside	The machinery — and so what should transfer to new cases
Engineering convenience	Nothing about similarity. The artificial system is a tractable stand-in for study	That the model is useful for the question being asked	Nothing about the natural system's mechanism. A claim about usefulness

The third is not a weaker version of the other two. It is a different kind of claim altogether.

A Criterion, and Its Most Famous Objection



From outside: fluent answers. Inside: nobody understands anything.

Searle's claim — the same is true of any computer running a program, however fluent its output.

Turing (1950). If you cannot tell the hidden partner from a person, the machine passes. The criterion is deliberately **behavioral**.

Searle (1980). The room answers fluently and no one inside understands a word. Behavior of the right kind is not sufficient for understanding.

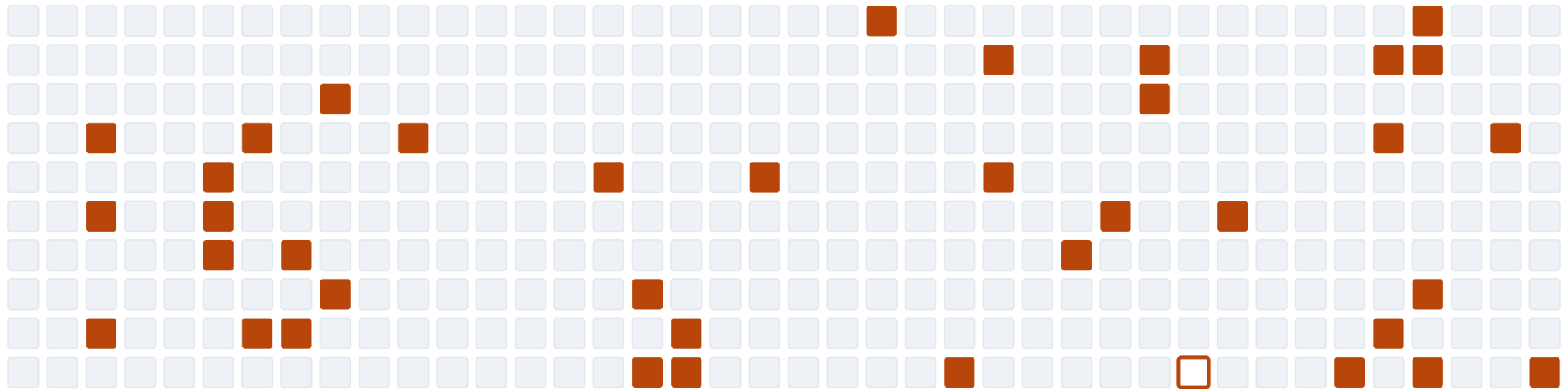
Both sides are reading machinery off a behavioral match — which is **surface matching**, the error we named in the first lecture of this module.

Remember Our Computer Vision Example?

In this module's first lecture: a person and an image model, each right about **95%** of the time, agreeing with each other **95%** of the time. Independence alone would already produce about 90.5%. I said the **surplus** is what needs explaining, and left it there.

The same 400 photographs from two weeks ago.

Then, the question was about the 362 that agree. Now it is about the 38 that do not.

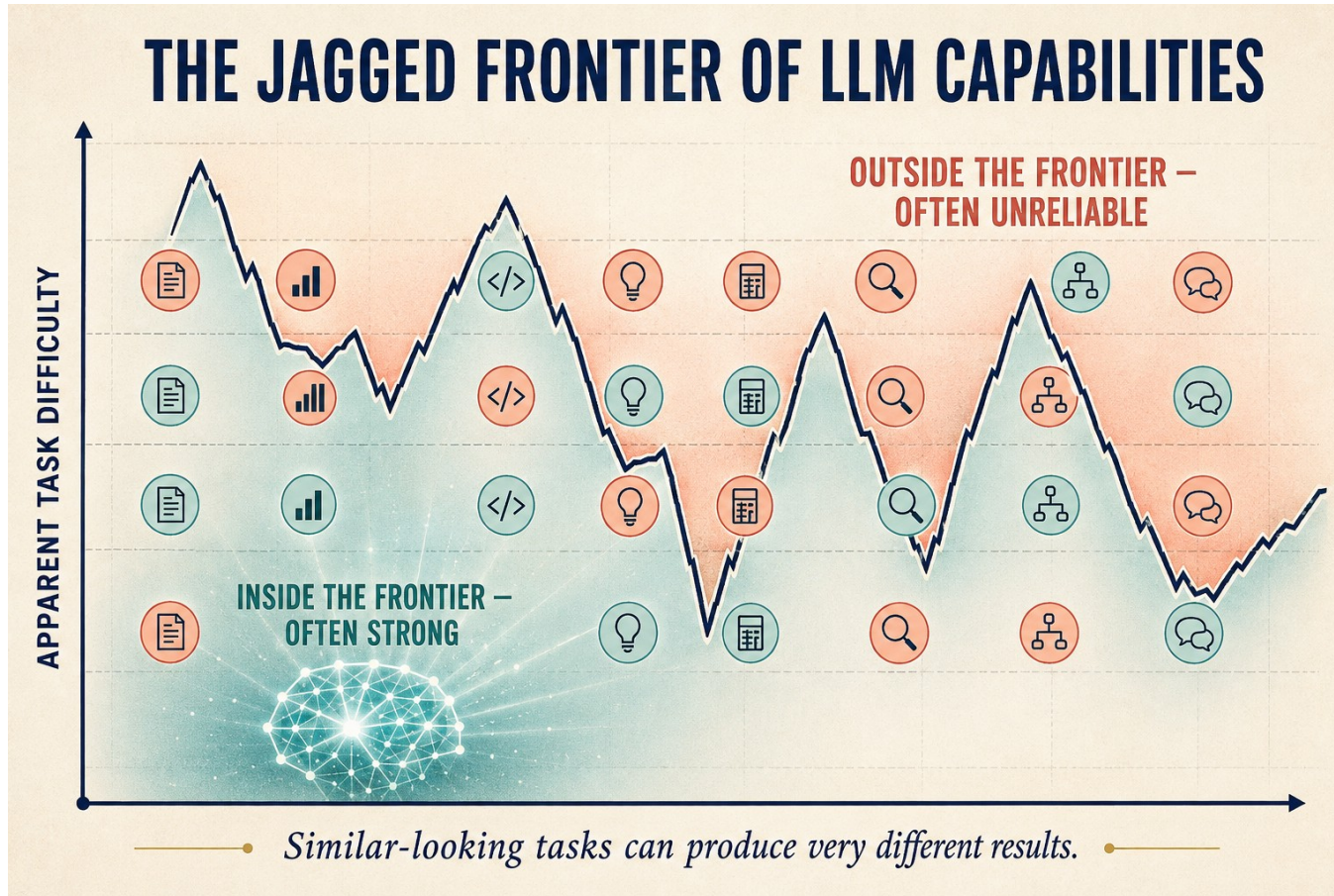


38 disagreements — and this is where the evidence is.

- Do they fail on the same photographs?
- Does noise degrade them along the same curve?
- In the same proportions, and in the same direction?
- Does each generalize to a new category the same way?

Agreement on the easy cases is cheap. Agreement on the failures is informative.

Run the Test on a Language Model — Then on Yourself



The verdict is mixed, and honestly mixed.

Behaviorally close · mechanistically uncertain ·
diverging under pressure.

Now the other direction. The common dismissal: it is *just* predicting the next word, so it is not really thinking.

That assumes two things. That we know what real thinking is at the mechanistic level. And that prediction is not it.

We do not know the mechanisms of human thinking. And an influential family of theories holds that the brain is itself largely a prediction engine — constantly anticipating its next input and learning from the error.

This anatomical engraving shows the human brain from a superior perspective. The two cerebral hemispheres are clearly visible, separated by the longitudinal fissure. The corpus callosum is shown in the center, with its various parts labeled. The cerebral cortex is highly convoluted, with numerous gyri and sulci. Various structures are labeled with letters: 'A' for the cerebral cortex, 'B' for the corpus callosum, 'C' for the corpus callosum, 'D' for the corpus callosum, 'E' for the corpus callosum, 'F' for the corpus callosum, 'G' for the corpus callosum, 'H' for the corpus callosum, 'I' for the corpus callosum, 'J' for the corpus callosum, 'K' for the corpus callosum, 'L' for the corpus callosum, 'M' for the corpus callosum, 'N' for the corpus callosum, 'O' for the corpus callosum, 'P' for the corpus callosum, 'Q' for the corpus callosum, 'R' for the corpus callosum, 'S' for the corpus callosum, 'T' for the corpus callosum, 'U' for the corpus callosum, 'V' for the corpus callosum, 'W' for the corpus callosum, 'X' for the corpus callosum, 'Y' for the corpus callosum, 'Z' for the corpus callosum.



Two things are true about it at once.

The fair one. Holding a belief firmly enough that it takes a lot to move you is not irrational — a more surprising claim *should* need stronger evidence. And when an argument arrives somewhere you are confident is wrong, you are entitled to suspect the argument.

So, what separates them is not how strongly the conviction is felt. It is whether you can say, in advance, what would change it.

Building One as a Way of Finding Out

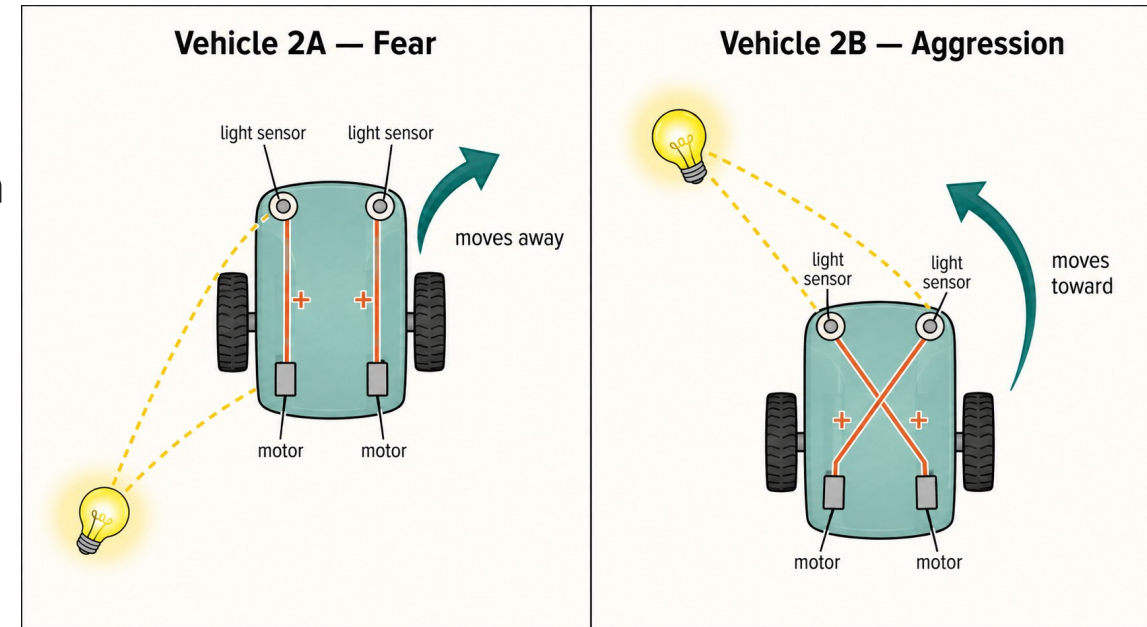
An artificial system does not have to be a candidate mind to be scientifically useful. **It can be a hypothesis.**

Building a system that reproduces some capacity is a way of finding out **what that capacity requires.**

And the failures are as informative as the successes: if a model has to be given something extra before it can handle a new situation, **that requirement is a claim about what the natural system needs too.**

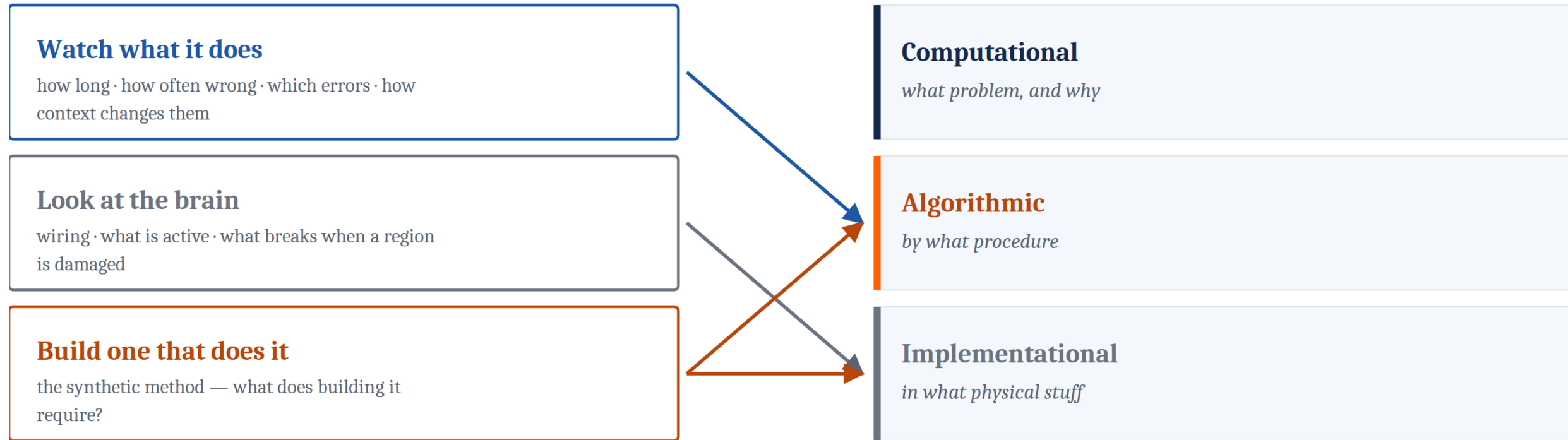
This has a name — the **synthetic method**, sometimes called understanding by building — and a long history in this field.

You have already done it. Two sensors, two motors, a couple of wires — **crossed or uncrossed**. And you described what you built as *afraid*.



Three Ways to Study a Mind

One mind. Three ways of looking at it — and each speaks to a different level.



Not three competing accounts of the same thing. Attempts on different levels of one explanation.

Now one mind on the table, and what gets compared is **the ways we have of looking at it.**

Every Method Has a Blind Spot

Three methods whose weaknesses have nothing to do with each other.

Behavior

Turn a face upside down and it becomes much harder to recognize. Do the same to a house and it costs far less.

blind spot: shows that something changed, not what changed inside

The brain

A patch of cortex under the temporal lobe responds far more strongly to faces than to other objects.

blind spot: shows where or when, never what it was accomplishing

Damage

Prosopagnosia: normal vision, can describe the eyes and nose, cannot tell whose face it is.

blind spot: shows a part is necessary, not how the whole works



Recognizing a face is not simply object recognition applied to faces.

No single technique's flaws could have produced agreement across all three. That is converging operations.

And When They Disagree

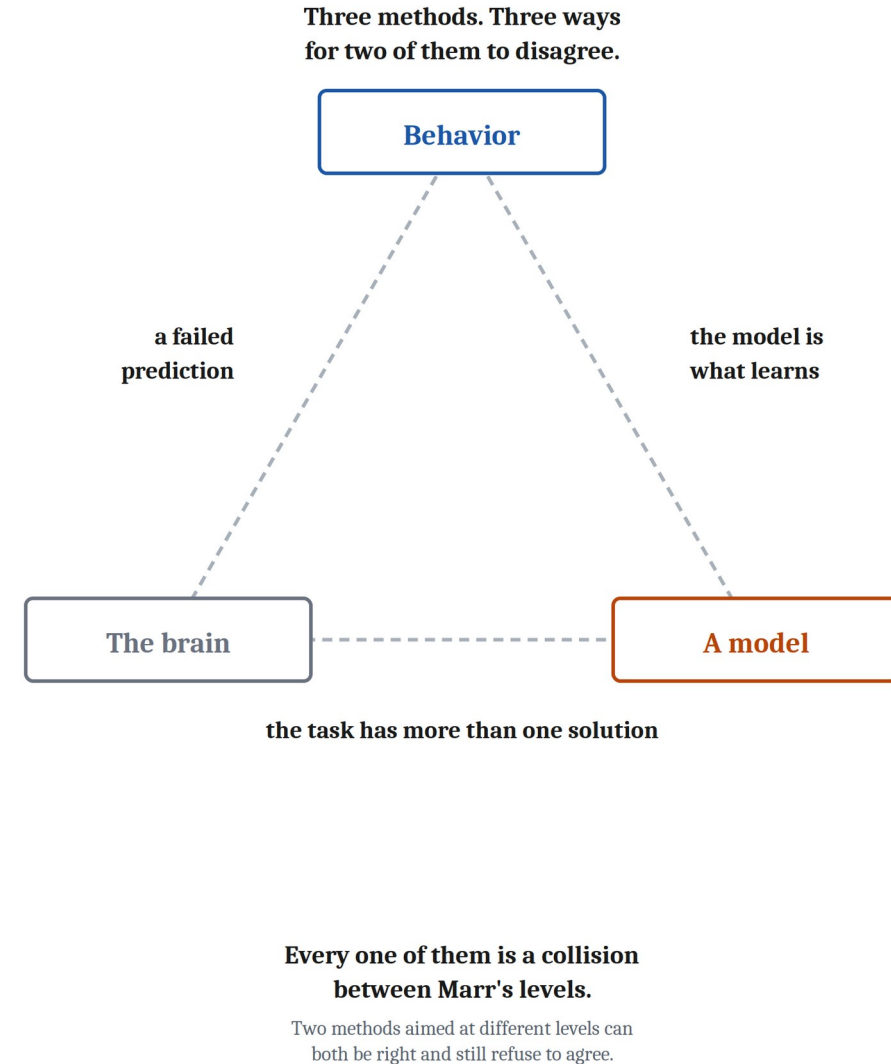
Agreement is not the only informative outcome. When two methods disagree, **the disagreement is itself a finding.**

And **which two disagree** tells us what kind of finding we have.

Brain and behavior. Primary visual cortex is where signals from the eye first arrive. Destroy it and vision in that part of the field should be gone — and patients report seeing nothing there. Ask them to guess where a light appeared and they say guessing is pointless. Then they guess **far better than chance.**

Behavior and a model — the lesson is usually about the model. A model that reproduces most of a behavior and breaks on one case is telling us precisely what it is missing.

Brain and a model — the lesson is often that the task can be solved more than one way, and the brain's particular solution is a fact needing its own explanation.



The Error This Family Makes

Reverse inference: going from "*this brain region was active during the task*" to "*therefore the task involved the process that region does.*"

An activation is a relationship between two measurements, and a relationship never interprets itself.

How much a given activation licenses depends on **how selectively that region responds** — an empirical question whose answer differs region by region.

How much an activation licenses depends on the region.

A SELECTIVE region

fear



Region X
is active

therefore the task involved ...



The backward step is reasonable.

One process drives the region, so activity is evidence of it.

Most regions are not like that.

A PROMISCUOUS region

fear

arousal

saliency

novelty

effort



Region X
is active

therefore the task involved ...



The backward step is weak.

Activity is consistent with all five — it does not pick one.

And the activation itself does not say which kind of region you have.

Two Threads, and What Comparison Provides

From Module 3 onward,
two paths at once.

From Module 3 onward, two paths at once

The evolutionary narrative

Natural vs. artificial comparisons

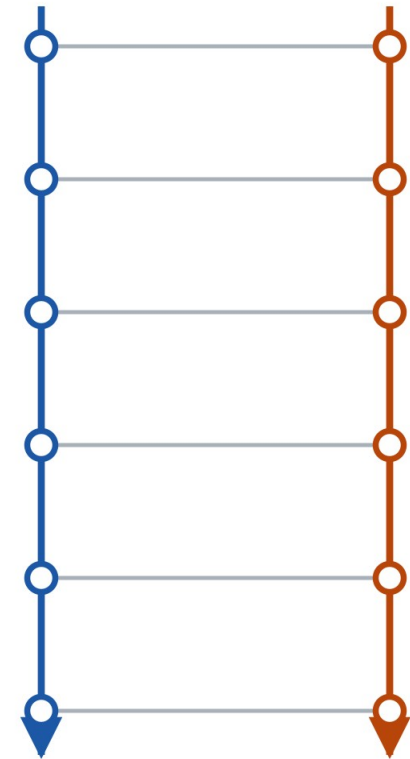
Each pairing is an instance of the **same three questions**: why do these two resemble each other, given that one was designed and one evolved • how deep does it go • how precisely can we state it.

Done carefully, comparison licenses conclusions about origins, about mechanisms, and about the problems a capacity was a solution to. Not bad for a method that requires only looking at two things at once.

What it does not do: settle what any creature is owed. And it cannot reach all the way into another mind — what it is truly like to be an octopus, an infant, or a machine is not a question this method answers.

The evolutionary
walk

An artificial
system



**Every rung is the same
three questions.**

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3	The three senses in which a machine can match a mind	fig-three-modes.png	CC BY 4.0	Jon Willits — drawn for this lecture from the Module 2 reading's table · own work
4	The Chinese Room	fig-chinese-room.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
5	The same 400 photographs, with the question moved to the disagreements	fig-breaking-same-way.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
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7	The base of the brain, from Vesalius's Fabrica (1543)	vesalius_base_of_brain.jpg	public domain	Andreas Vesalius, De humani corporis fabrica (1543); scan via e-rara · https://commons.wikimedia.org/wiki/File:1543,_Andreas_Vesalius_%27_Fabrica,_Base_Of_The_Brain.jpg
7	A current laptop	macbook_pro.jpg	CC BY-SA 4.0	Kyu3a, own work — CC BY-SA 4.0 · https://commons.wikimedia.org/wiki/File:M3_Macbook_Pro_14_inch_Space_Grey_model_(cropped).jpg
8	Braitenberg's Vehicle 2, wired both ways	braitenberg_vehicles.png	Jon's own generated image; cleared for classroom and web use	Jon Willits — generated with ChatGPT · own work

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9	The three methods, and the levels each one speaks to	fig-three-methods.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
10	Three methods with almost nothing in common, converging on one conclusion	fig-converging-faces.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
11	Three methods, and the three ways two of them can disagree	fig-disagreement-triangle.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
12	The backward step, for a selective region and a promiscuous one	fig-reverse-inference.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work
13	Two rails, joined at every step	fig-two-threads.png	CC BY 4.0	Jon Willits — drawn for this lecture · own work